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Introduction to Algorithms and Data Structures, Exercise Sheet 13

https://cms.sic.saarland/gralgodat25/

Total Points: **50 bonus** Due: 12:00 Thursday, **January 29**, 2026

You are allowed to collaborate on the exercise sheets in groups of up to 3 students. Each group must submit exactly one solution, that is, exactly one group member should submit the solutions. The first page of your solutions must list all group members and the name of your respective tutor. Further, cite all external sources that you use (books, websites, research papers, etc.). You need to collect at least 50% of all points on exercise sheets to be admitted to the exam. Upload your solutions in PDF format directly to the CMS, or hand them in physically at the beginning of the lecture.

Please note: An exercise that asks you to design or implement an algorithm is to be answered by (1) your algorithm in pseudocode, (2) a correctness proof, and (3) a running time analysis.

Exercise 1 (Ambitious Student Revisited) — 16 bonus points

Recall the Ambitious Student problem from the lecture: Given intervals $L_1 = [s_1, e_1), \ldots, L_n = [s_n, e_n)$, compute the largest number k such that there is a conflict-free set of k intervals $L_{i_1}, \ldots, L_{i_k}$. You have seen a greedy algorithm that solves the problem by picking the interval that *ends first*. In this exercise, we will investigate further strategies; your task is to decide for each one whether it is correct or not. If the strategy is correct, then give a short proof (you don't have to analyze the running time). Otherwise, give a counterexample.

a. Pick the interval L_i that *starts first*, discard all intervals that conflict with L_i and recurse.

b. Pick the interval L_i that *starts last*, discard all intervals that conflict with L_i and recurse.

c. Pick the interval L_i that *conflicts with the fewest other intervals*, discard all intervals that conflict with L_i and recurse.

d. If no intervals conflict, pick them all. Otherwise, discard the *longest* interval and recurse.

e. If no intervals conflict, pick them all. Otherwise, discard the interval that *conflicts with the most other intervals* and recurse.

f. Let L_i be the interval with the *earliest start time*, and let L_j be the interval with the *second earliest start time*.

- If $L_i \cap L_j = \emptyset$, pick L_i and recurse on everything but L_i.
- If $L_j \subseteq L_i$, discard L_i and recurse.
- Otherwise, discard L_j and recurse.

g. If any interval L_i completely contains another interval, discard L_i and recurse. Otherwise, pick the interval L_j that *starts first*, discard all intervals that conflict with L_j and recurse.

Hint: exactly three of these strategies are correct.

Exercise 2 (Optimal Interval) — 8 bonus points

Design an algorithm that solves the following task in time $O(n)$: Given an array $A[1 \ldots n]$ of (possibly negative) integers, find two indices $i \leq j$ such that $\sum_{k=i}^{j} A[k]$ is as large as possible.

Exercise 3 (Tale of Mr. Clerk) — 9 bonus points

Mr. Clerk receives a pile of n jobs, where every job j has a *deadline* $D_j \in \mathbb{N}$. On each day $t = 1, 2, \ldots,$ he picks one job and removes it from the pile. But he may only pick a job whose deadline had not passed; that is, a job j satisfying $t \leq D_j$. If no such job exists, then he gets fired. (He will be fired on day n at the latest.) Naturally, he wants to maximize the number of days he stays.

Design an $O(n \log n)$-time algorithm that, given $D_1, \ldots, D_n$, computes the maximum number of days Mr. Clerk can stay.

Exercise 4 (Watchmen) — 9 bonus points

An art musuem is recruiting night watchmen and soon receives n applications. Each applicant i specifies his/her availability as an interval $[a_i, b_i]$; for example $[20.5, 27]$ means that the applicant is available from 8:30pm to 3am the next day.

The museum must be fully secured from 6pm to 9am the next day, meaning that at least one watchman is present at any moment. Design an algorithm that, given real numbers $18 \leq a_i < b_i \leq 33$ for each i, computes the minimum number of watchmen the museum should hire. You will get partial points if your algorithm runs in $O(n^2)$ time, and full points if it runs in $O(n \log n)$ time.

Exercise 5 (Broadcasting) — 4+4 bonus points

A message is to be broadcast to all n nodes in a given rooted tree. Initially, only the root node knows the message. In each round, every node that knows the message can forward it to at most one of its children. We want to compute the minimum number of rounds required to broadcast the message to all nodes.

a. Assume the input is a k-ary tree where k is a constant. (See Figure 1 for an example when $k = 2$.) Design an algorithm that solves the problem in time $O(n)$.

b. For general input trees, design an efficient algorithm that solves the problem. (You will get partial points if your algorithm runs in polynomial time, and full points if it runs in time $O(n)$.)

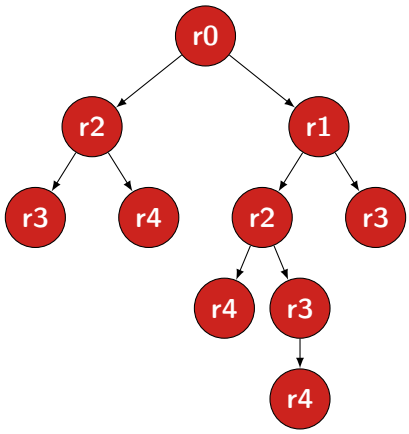

Figure 1. An example of the broadcasting problem where the input is a binary tree. It requires four rounds to broadcast the message to all nodes. The labels indicate in which round the respective node has learnt the message in an optimal solution.